

AI-Based Resume Screening and Candidate Ranking System

Introduction

Recruitment is an important process in every organization. When a company receives a large number of resumes for a job opening, manually reviewing each resume becomes time-consuming and inefficient. This project aims to simplify the recruitment process by using Artificial Intelligence and Natural Language Processing techniques to automatically analyze resumes and compare them with job requirements.

The system helps recruiters identify suitable candidates by ranking resumes based on their relevance to a given job description. It also highlights matching skills, missing skills, and provides a recommendation to assist in the hiring process.

Abstract

The AI-Based Resume Screening and Candidate Ranking System is designed to automate the initial stage of candidate selection. The system accepts a job description and multiple candidate resumes in PDF format. Resume text is extracted and processed using NLP techniques. Sentence Transformer embeddings are generated for both job descriptions and resumes, and semantic similarity is calculated to determine how closely a candidate matches the job requirements.

The system ranks all candidates according to their scores and provides additional information such as matched skills, missing skills, skill match percentage, and hiring recommendations. A Streamlit-based interface allows recruiters to upload resumes, view rankings, and download reports easily.

Tools Used

- Python
- Streamlit
- Sentence Transformers
- spaCy
- Scikit-learn
- Pandas
- NumPy
- PyMuPDF
- PyTorch
- Visual Studio Code

Steps Involved in Building the Project

1. Data Collection

A resume dataset containing resumes from different job categories was collected and analyzed.

2. Data Preprocessing

Resume text was cleaned by removing unwanted characters, extra spaces, and formatting issues. This helped improve text quality for further processing.

3. Resume Parsing

PDF resumes were processed using PyMuPDF to extract text content automatically.

4. Job Description Analysis

The system extracts important information such as required skills, education, and experience from the entered job description.

5. Semantic Similarity Calculation

Sentence Transformer embeddings were generated for resumes and job descriptions. Cosine similarity was used to calculate the relevance score between them.

6. Skill Matching

The extracted skills from the job description were compared with the resume content to identify matched and missing skills.

7. Candidate Ranking

Candidates were ranked according to their overall score and skill match percentage.

8. Frontend Development

A Streamlit-based dashboard was developed to allow users to upload multiple resumes, analyze candidates, and view ranking results interactively.

Conclusion

The AI-Based Resume Screening and Candidate Ranking System successfully automates the initial recruitment process. The system reduces manual effort by analyzing resumes, matching them with job requirements, and ranking candidates automatically. It provides recruiters with useful insights such as skill matching, missing skills, and hiring recommendations. The project demonstrates the practical application of Artificial Intelligence and Natural Language Processing in modern recruitment systems and can be further enhanced with OCR support, LinkedIn integration, and advanced recommendation features.